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A Synopsis on

_____Title of project_____

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Proposed to be submitted in

partial fulfilment of the requirements for the award of degree of

Bachelor of Engineering

By

Student 1 name	Roll Number
Student 2 name	Roll Number
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Under the guidance of

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DataLens

A Web-Based Platform for Exploratory Data Analysis and Dataset Readiness Assessment

1. Abstract

Organizations and student projects often receive raw tabular data before its structure, quality, and analytical usefulness are understood. Examining a dataset manually requires repeated checks across columns, missing entries, duplicated records, distributions, relationships, and possible preprocessing needs. DataLens is a browser-based exploratory data analysis (EDA) platform developed to bring these first investigative steps into one workflow. A user uploads a comma-separated values (CSV) file and can review its dimensions, column data types, missingness, duplicate rows, distinct-value counts, numerical summaries, and categorical summaries. The application adds data-quality checks for outliers, constant columns, and high-cardinality fields; descriptive and distribution statistics; correlation analysis; and automatically generated charts. An optional target column enables target distribution and feature-relationship plots. A feature-engineering advisory module suggests candidate strategies for missing values, categorical encoding, scaling, skewness transformation, correlated features, and class imbalance. The results are assembled into a consolidated, browser-viewable analysis report. DataLens is implemented with Python, FastAPI, pandas, NumPy, Matplotlib, Seaborn, Jinja2, HTML, CSS, and JavaScript. Its recommendations are heuristic aids for analyst review: the current system does not automatically modify the uploaded data or train predictive models. The project aims to make early-stage data understanding faster, more consistent, and accessible through a single web interface.

Keywords: exploratory data analysis, data profiling, data quality, feature engineering, visualization, CSV, FastAPI.

2. Introduction

Data analysis begins with understanding what a dataset contains and whether it is suitable for the question being studied. Real-world tables may contain incomplete records, duplicated observations, inconsistent data types, skewed numerical distributions, unusual values, redundant variables, and imbalanced target classes. These conditions can influence later statistical conclusions and machine-learning preparation. They are not always obvious from a spreadsheet preview, particularly when the dataset has many columns or a large number of records.

Exploratory data analysis combines numerical summaries and visual inspection to reveal structure, patterns, and potential anomalies before formal modelling. In practice, analysts often repeat the same profiling, quality, statistical, and plotting operations for every new dataset. The results may be scattered across scripts or notebooks, making it harder to compare findings or communicate them to collaborators. A web application that collects these steps in a repeatable sequence can reduce this friction while keeping the analyst responsible for interpreting the evidence.

DataLens addresses this need through a modular web workflow. FastAPI receives file uploads and routes requests to analysis services. pandas reads CSV content into tabular data structures, while NumPy supports numerical computations. Dedicated services compute profiles, quality indicators, statistical findings, feature-preparation recommendations, and visualizations. Generated results are stored as analysis sections and displayed in a Jinja2-based report interface. The modular design makes the main analytical responsibilities easier to inspect and extend.

3. Problem Statement

Users who need to assess a new tabular dataset must commonly run separate commands or scripts to understand its schema, count missing and duplicate values, inspect numerical and categorical distributions, identify potential outliers or redundant features, and create visual summaries. This fragmented process takes time, depends on the user remembering which checks to perform, and can lead to important data-quality concerns being overlooked or findings being difficult to communicate. There is a need for an accessible application that accepts a CSV dataset, performs a consistent first-pass analysis, presents understandable evidence, and offers transparent preprocessing suggestions without making unreviewed changes to the source data.

4. Aim and Objectives

• To design and implement a browser-based tool that enables users to upload a CSV dataset and explore it through a unified interface.

• To summarize dataset dimensions, column types, missing-value counts and percentages, duplicate rows, unique values, and numerical and categorical characteristics.

• To detect and present common data-quality signals, including missingness, duplicate observations, constant columns, potential numerical outliers, and high-cardinality fields.

• To compute descriptive statistics, categorical frequency summaries, distribution shape measures, strong numerical correlations, and rule-based textual insights.

• To generate visual summaries for numerical and categorical columns, missingness, correlations, and optional relationships with a user-selected target variable.

• To provide per-feature preprocessing recommendations for imputation, encoding, scaling, skewness transformation, high-cardinality and constant features, strong correlations, and target-class imbalance.

• To consolidate available analysis outputs into a readable report that supports review and communication of initial findings.

• To evaluate the application on representative CSV datasets by checking workflow completion, consistency of reported measures, and clarity of the generated visual and textual outputs.

5. Scope of the Project

The current scope is exploratory analysis of tabular data supplied as CSV files. The user starts an analysis by uploading a dataset through the web interface. Profiling reports the file name, row and column counts, duplicate-row count, per-column data type, missing count and percentage, and number of distinct values. Numerical columns receive summaries such as mean, median, minimum, maximum, and standard deviation. Categorical columns receive their most frequent value and distinct-value count.

The quality-analysis component reports missing fields, duplicate rows, constant columns, potential outliers using the 1.5 × interquartile-range rule, and high-cardinality columns. Statistical analysis adds numerical measures including mean, median, mode, range, variance, standard deviation, skewness, and kurtosis; categorical frequencies; pairwise numerical correlations; and rule-based observations about skewness, correlation, outliers, missingness, and constant columns. Thresholds used by the current implementation are screening rules and should be interpreted in context, not as proof that a value or variable is erroneous.

The visualization component generates histograms and box plots for numerical columns, count plots and pie charts for categorical columns, a numerical correlation heatmap, and a missing-value plot. If the user supplies a target column, DataLens also creates target-distribution and target-versus-feature visualizations for supported numerical and categorical fields. The report page presents the stored sections together, including the available insights, feature-engineering

suggestions, and charts.

Feature-engineering output is advisory. Current recommendations include median or most-frequent imputation guidance; one-hot, label, or ordinal encoding suggestions; StandardScaler or RobustScaler selection based on an outlier heuristic; log or Yeo-Johnson transformation suggestions based on skewness and negative values; warnings for high-cardinality, constant or near-constant fields; suggestions for highly correlated features; and class-imbalance guidance when a target is provided. The project does not currently apply these transformations to the source table, train a model, guarantee that an observation is erroneous, or establish causal relationships.

The implemented upload and analysis path is designed for CSV input. Support for Excel, JSON, databases, very large streaming datasets, user accounts, persistent multi-user workspaces, downloadable reports, automated cleaning, and predictive modelling is outside the current implemented scope and can be considered as future work. The current analysis store writes sections to a local JSON file; deployment for simultaneous multi-user use would require a session-aware or database-backed storage design.

6. Proposed System and Methodology

6.1 System architecture

Presentation layer: HTML templates, CSS, and JavaScript provide the upload and report interfaces. **Request layer:** FastAPI endpoints receive uploaded files and, where required, an optional target-column value. **Analysis layer:** separate Python services perform profiling, quality checks, statistical analysis, feature-engineering recommendation generation, and plotting. **Persistence and reporting layer:** analysis sections are saved in a local JSON store and loaded to populate the consolidated report. This separation allows individual analysis modules to be refined without placing all calculations in the interface layer.

6.2 Analysis workflow

• **Upload and parse:** receive the user-selected CSV file and read it into a pandas DataFrame.

• **Profile:** determine dataset dimensions, duplicate count, column types, missingness, unique counts, numerical summaries, and categorical summaries.

• **Assess quality:** identify missing and duplicate data, constant columns, IQR-based numerical outliers, and columns whose unique-value count exceeds the implemented cardinality threshold.

• **Analyze statistics:** calculate descriptive and categorical statistics, distributions, pairwise numerical correlations, and rule-based insights.

• **Recommend preparation:** apply documented threshold rules to produce feature-level suggestions; present them as review prompts rather than executing changes.

• **Visualize and report:** generate plots, optionally compare fields with a target column, save analysis outputs, and render the consolidated report.

6.3 Main modules

Module	Input and key output
Dataset profiling	CSV table; dimensions, schema, missingness, duplicates, numerical and categorical summaries.
Data quality	DataFrame; missing and duplicate counts, constant columns, IQR outlier counts, high-cardinality indicators.
Statistics and insights	DataFrame; descriptive statistics, category frequencies, correlations, skewness/kurtosis, rule-based observations.
Feature-engineering advice	DataFrame and optional target; heuristic preparation suggestions by feature.
Visualization	DataFrame and optional target; distribution, categorical, missingness, correlation, and target plots.
Consolidated report	Stored analysis sections; browser-rendered summary and visual output.

6.4 Proposed evaluation

Evaluation will use several CSV files containing a mixture of numerical and categorical variables, controlled missing values, duplicate rows, constant fields, and known category distributions. For each case, expected counts and summaries will be calculated independently and compared with the application output. Visual outputs will be checked for successful generation, correct labels, and correspondence with the selected columns. The report will be reviewed for completeness and readability. This evaluation assesses the reliability and usability of the first-pass analysis; it does not establish that heuristic recommendations are universally optimal.

7. Applications and Expected Benefits

DataLens can support students learning data analysis, researchers inspecting a new experimental table, analysts conducting an initial data audit, and small teams that need a quick overview before deciding how to prepare a dataset. It can help reveal where missingness is concentrated, which variables have unusual distributions, which numerical fields move together, whether an optional target is imbalanced, and what preparation questions should be investigated next. A consolidated report can make these initial findings easier to discuss and use as a starting point for more rigorous analysis.

The intended benefit is reduced repetitive setup and a more consistent first-pass checklist. It does not replace domain expertise: an outlier may be valid, a correlated feature may be important, and an imputation or encoding choice depends on the analytical objective. Users should inspect the underlying data and select any preparation method based on context.

8. Project Plan and Timeline

The following schedule is proposed for the 2026–2027 academic year. It organizes the project into review, implementation, evaluation, and documentation phases; the month labels should be aligned with the institute calendar and guide’s review milestones.

Phase	Planned activities	Indicative period	Deliverable
1. Requirements and study	Review EDA, data quality, preprocessing guidance, visualization, and web application approaches; finalize requirements and representative datasets.	Months 1–2	Approved requirements and synopsis refinement
2. Design	Define modules, upload/report workflow, endpoint responsibilities, storage approach, and evaluation criteria.	Months 2–3	System design and test dataset plan
3. Core implementation	Implement upload, CSV parsing, profiling, data-quality checks, statistical analysis, and insight generation.	Months 3–5	Working analysis modules

Phase	Planned activities	Indicative period	Deliverable
4. Visual analysis and report	Complete plots, optional target analysis, feature-preparation recommendations, and consolidated report presentation.	Months 5–7	Integrated DataLens application
5. Verification and refinement	Compare outputs with independently calculated results; check representative edge cases; improve error handling and presentation.	Months 7–9	Evaluation results and refined prototype
6. Final documentation	Prepare technical documentation, final synopsis/report, demonstration material, and project submission.	Months 9–10	Final report and presentation

9. Limitations and Future Enhancements

Current rules use fixed thresholds, so they provide screening signals rather than context-aware judgments. CSV parsing and analysis are performed through the implemented request workflow, and the local JSON analysis store is not a multi-user database. Report access depends on the application environment and currently stored analysis. Future versions may add robust input validation and error messages, configurable thresholds, session-isolated storage, more file formats, exportable reports, larger-data processing, user-selected cleaning pipelines with before-and-after previews, and carefully scoped predictive modelling. These extensions should preserve clear user control and explain the assumptions behind each result.

10. Conclusion

DataLens brings core exploratory data analysis tasks into a single web application for CSV datasets. It combines dataset profiling, data-quality indicators, descriptive and categorical statistics, correlation and distribution analysis, visual summaries, heuristic preprocessing recommendations, and a consolidated report. Its modular FastAPI and Python service structure provides a practical foundation for iterative development. By making first-pass evidence easier to produce and review, the project can help users decide what to investigate next while leaving data-cleaning and modelling decisions under analyst control. The next stage is systematic verification across representative datasets, followed by usability and reliability improvements informed by evaluation.

11. References

[1] FastAPI. “Request Files.” FastAPI Documentation. https://fastapi.tiangolo.com/tutorial/request-files/ (accessed 24 September 2026).

[2] pandas development team. “pandas.read_csv.” pandas Documentation. https://pandas.pydata.org/docs/reference/api/pandas.read_csv.html (accessed 24 September 2026).

[3] Harris, C. R., et al. “Array programming with NumPy.” Nature, vol. 585, pp. 357–362, 2020. https://doi.org/10.1038/s41586-020-2649-2.

[4] Hunter, J. D. “Matplotlib: A 2D Graphics Environment.” Computing in Science & Engineering, vol. 9, no. 3, pp. 90–95, 2007. https://doi.org/10.1109/MCSE.2007.55.

[5] Waskom, M. L. “seaborn: statistical data visualization.” Journal of Open Source Software, vol. 6, no. 60, p. 3021, 2021. https://doi.org/10.21105/joss.03021.

[6] scikit-learn developers. “Preprocessing data.” scikit-learn User Guide. https://scikit-learn.org/stable/modules/preprocessing.html (accessed 24 September 2026). Used as background for common preprocessing concepts; DataLens provides its own rule-based suggestions.

[7] pandas development team. “DataFrame.describe.” pandas Documentation. https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.describe.html (accessed 24 September 2026).

Synopsis Approval

This synopsis presents the proposed Bachelor of Engineering project entitled **“DataLens: A Web-Based Platform for Exploratory Data Analysis and Dataset Readiness Assessment.”** The undersigned acknowledge review of the project proposal for the Department of Artificial Intelligence and Data Science.

Place: _____ Date: _____

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